Task-Oriented Semantic Granularity Selection for Bandwidth-Constrained V2V Cooperative Perception

Peiyi Yue

Abstract—Cooperative perception must choose what to transmit, not only how much to compress. We formalise per-frame semantic granularity selection over three actions—ego-only (E), object-level (L) and feature-level (F)—and evaluate it under a single detector, a single field of view and a single ground truth, with communication charged through a measured LDPC/QAM chain rather than a declared budget. The resulting method, CA-TOSG (channel-aware task-oriented semantic granularity), reduced realised communication payload by 99.85% on Test and 99.70% on Culver-City relative to the frozen comparator. However, the preregistered scene-level non-inferiority criterion was not met on either split, indicating that substantial communication reduction was achieved without statistically establishing accuracy preservation. Three mechanism results follow from the unified branch: partial recovery is necessary, since all-or-nothing delivery of a $12,567$-codeword message is effectively dead at realistic loss rates; where a loss falls changes the task outcome even when how much is lost is held exactly fixed; and fusing a zeroed feature tensor is not equivalent to ego-only inference. In the frozen policy the feature action is never selected, and we show this follows from the cost structure and the sampled penalty grid rather than from a failure to learn.

Index Terms—Vehicle-to-vehicle communication, cooperative perception, semantic communication, bandwidth adaptation, 3D object detection.

I. Introduction

CONNECTED and automated vehicles can exchange perception information over vehicle-to-vehicle (V2V) links for complementary viewpoints, which matters most where a single vehicle's sensing is incomplete—intersections, dense traffic, occlusions, long range—improving detection robustness beyond one vehicle's range and viewpoint [1], [2].

It is, however, constrained by communication. Raw point clouds and dense feature maps preserve the most information but can exceed what vehicular links carry under bandwidth and latency limits, and V2V channels are time-varying through mobility, blockage, multipath and fading [3], [4]: a strategy that works on a clean channel can be unreliable at low signal-to-noise ratio (SNR). Cooperative perception must therefore account for the reliability and cost of the link, not only for perception.

Existing methods sit at different points of that trade-off. Late fusion sends object-level predictions [1]—compact and protectable, but discarding semantic context—while intermediate fusion sends feature maps [2], [5], richer at much higher cost, which recent work reduces by region selection, masking or compression [5] and which semantic communication optimises for task performance [6], [7]. Prior work has considered multilevel or adaptive sharing, while CA-TOSG focuses specifically on receiver-driven per-frame E/L/F selection using strictly ego-local pre-request cues and a common digital transport-accounting protocol. On some frames local perception is already confident enough for the compact message; on occluded or ambiguous ones the richer message helps. The best granularity depends on the task state.

The channel decides the same question from the other side. A denser feature message needs a more reliable link, so under poor SNR or fading it can arrive damaged and its payload is partly wasted, while the much smaller object-level message is robust low-rate information. Richer semantic communication therefore does not always mean better perception; granularity depends jointly on task cues and channel quality.

This paper treats the choice itself as the object of study. CA-TOSG lets the ego receiver select one message granularity per frame from $\mathcal{S} = \{E, L, F\}$ using strictly ego-local cues and an estimated channel state, and signal it with a two-bit request; the collaborator complies (Fig. 1). Our contributions are three.

- Problem formalisation. We pose per-frame granularity selection as a budget-constrained action choice over $\{E, L, F\}$, with payload charged per frame through the transport chain the messages actually traverse rather than through a declared constant. The formulation is the Lagrangian relaxation of a constrained perception–communication problem (Section III-G).
- A receiver-driven method. The ego evaluates its own scene and channel and requests a granularity. Every selector input is ego-local and available before the request is issued, so no input depends on a message the ego has not yet asked for—which removes the causality problem that a sender-side policy would face.
- Experimental findings under a unified protocol. One checkpoint, one field of view, one ground truth and a deterministic evaluation pipeline. Under that protocol we report a large communication reduction whose accuracy non-inferiority is not established, together with three mechanism results—the necessity of partial recovery, the effect of loss position at fixed loss amount, and the inequivalence of zero-tensor fusion and ego-only inference—and an account of why the frozen policy never selects the feature action.

The random forest used below is one instance of the selection layer, not the contribution. What is offered is the granularity-control formulation and the accounting protocol that makes a payload claim checkable; any policy that consumes ego-local cues and emits a request slots into the same framework, and we report below that this particular instance does not beat a tuned threshold at equal budget.

II. Related Work

A. V2V Cooperative Perception

V2V cooperative perception exploits information from surrounding connected vehicles to alleviate occlusion, long-range sparsity and limited field of view [1]. Methods are categorised by the stage at which information is shared. Early fusion shares raw point clouds before feature extraction [1]: the most complete information, but a very large payload requiring accurate spatio-temporal alignment. Late fusion shares object-level predictions—boxes, classes, confidences [1]. The payload is small and suits low-rate links, but intermediate semantic and spatial information is discarded: an object missed by the sender cannot be recovered by the receiver. Intermediate fusion shares learned bird's-eye-view (BEV) features between backbone and detection head [2], [5], [8], [9], balancing the two extremes at a much larger message size. Benchmarks such as OPV2V and OpenCOOD [1] provide common datasets and protocols for comparing these strategies.

Relation to CA-TOSG. These works fix the representation and study fusion within it. CA-TOSG keeps all three representations available and decides, per frame, which one this frame should use.

B. Communication-Efficient Cooperative Perception

To reduce the communication burden of intermediate fusion, many methods transmit only task-relevant features, using confidence maps, attention weights or importance masks to select informative spatial regions [5], [6], [10], [11]. Where2comm [5] is representative: it estimates where communication is needed and selectively transmits spatial feature regions. Feature compression is a second direction, encoding the map into a lower-dimensional representation before transmission [6], [12]. ML-Cooper [13] adapts the proportion of raw-, feature- and object-level data to the network state with a soft actor–critic policy, and RBH [14] chooses hybrid fusion by spatial overlap.

Relation to CA-TOSG. Where2comm selects which spatial positions to send inside the feature level; importance-map and JSCC methods optimise how the feature map is encoded and carried over the channel; SmartCooper adjusts the feature compression ratio. ML-Cooper is the closest in spirit, in that it mixes levels, but it does so as a heavyweight learned resource-sharing policy with a different action mechanism. CA-TOSG decides which semantic granularity this frame uses at all, one level above the question those methods answer, and any of them can serve as the feature branch that CA-TOSG activates.

C. Semantic and Channel-Aware Communication for Vehicular Perception

Semantic communication optimises the transmitted representation for task performance rather than for reconstruction fidelity or bit error rate [6], [7], [15]–[17]. Joint source–channel coding trains encoder and decoder with the channel in the loop and can degrade more gracefully than separated source and channel coding, whereas digital QAM+LDPC transport shows threshold behaviour [3]: poor below an SNR and rapidly reliable above it. That dependence matters most for vehicular links, whose SNR varies with multipath, blockage and fading [4]. Some work makes the transmission channel-adaptive at a fixed granularity: SmartCooper [12] adjusts the compression ratio from the estimated channel state, and AccBEV [18] conditions a feature-repair module on the SNR. CoDS [19] pairs a task-oriented semantic codec with a semantic analog-to-digital converter so that learned features traverse a standard digital V2X bitstream, and discards features corrupted by decoding failures at the receiver.

Relation to CA-TOSG. All of these adapt the encoding, the repair or the rate within feature-level transmission. CA-TOSG acts before that: it decides whether a feature message is requested at all, and otherwise falls back to the compact object-level message or to no cooperative payload. The two are complementary—a digital semantic codec can be the feature branch that CA-TOSG activates.

III. System Model and Problem Formulation

A. Cooperative Perception Setting

We consider a single ego–collaborator V2V link. At frame t the ego observes the scene with its own LiDAR and extracts a local BEV feature $F_{\text{ego},t}$ with a PointPillars backbone [20]; the collaborator observes from its own viewpoint and extracts $F_{j,t}$ with the same checkpoint. The ego fuses whatever it receives with its local feature and produces the final 3D detection $\hat{Y}_t$, which is scored against the ground truth Y_t.

We adopt a receiver-driven architecture: the ego evaluates its own perception and channel state, decides which granularity it needs, and signals the choice with a 2-bit request. We treat the request as an application-layer extension associated with the ETSI Cooperative Awareness Message [21] and Collective Perception Message [22] services, carried over IEEE 802.11bd or NR sidelink [3]; the standards integration itself is outside the scope of this paper. The request is a fixed control overhead, identical for all three actions, and is therefore not charged to any action's payload. It is sent on every frame, including when the ego selects E; what E removes is the cooperative perception payload, not the request.

B. Message Candidates

The ego selects one action per frame,

$$s_t \in \mathcal{S} = \{E, L, F\}, \tag{1}$$

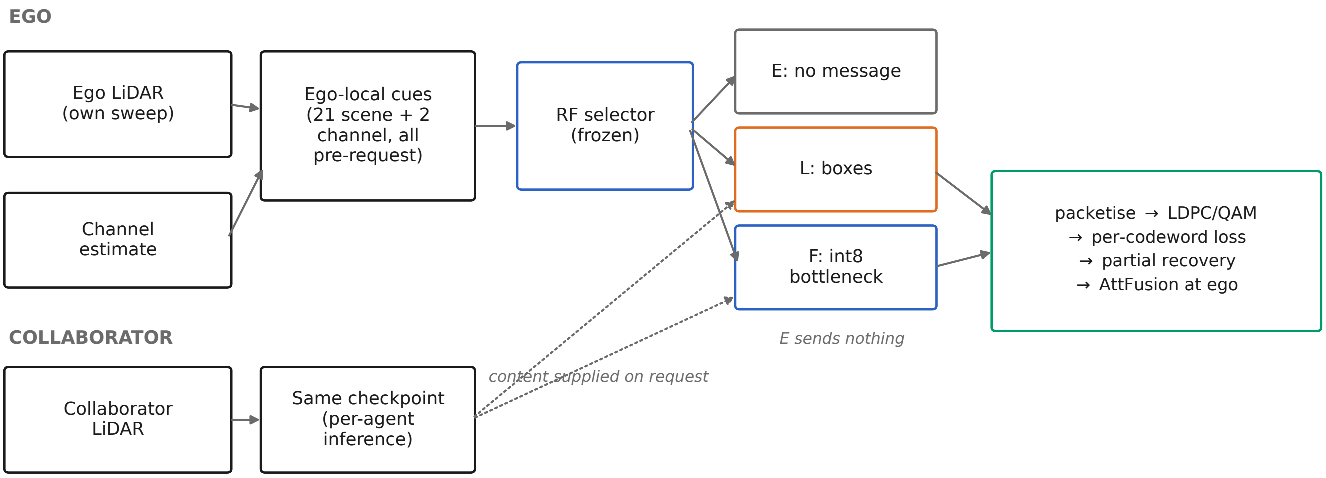

Fig. 1. Data flow as implemented. The cue vector and the channel estimate are ego-side and available before the request; the collaborator supplies message content only when asked. E issues no message and therefore never enters the transport chain.

where E is ego-only operation—the collaborator transmits no cooperative perception payload and the ego detects from its own LiDAR—L is the collaborator's detected boxes, and F is the collaborator's quantised bottleneck feature tensor.

E is a first-class action, not merely a failure mode. It costs no perception payload and cannot fail in transit, which is right both where the ego's own detection already suffices and where the channel is too poor for a message to be worth its cost.

The object-level message is a compact set of local detections, $m_t^L = \{(b_k, c_k, p_k)\}_{k=1}^{N_t}$, with b_k the 3D box, c_k the class and p_k the confidence. The feature-level message is the encoder's bottleneck tensor, $739,200$ elements, quantised to $w = 8$ bits per element before transmission. No modulation-order variant is part of the action space: the selector chooses what to send, whereas the constellation is a transport parameter of the link.

C. Channel Model and Transport

Let γ_t be the estimated channel quality at frame t in dB, available from pilot-based estimation or link-layer reports as already produced by IEEE 802.11bd and 5G NR sidelink stacks [3]. Messages are carried by a rate-1/2 LDPC code with information block $K = 500$ and codeword length $n = 1000$, modulated with 16-QAM.

A message is fragmented into packets of $8,000$ payload bits with a 320-bit header following common V2X framing assumptions [3], [4], and each packet is coded into LDPC codewords. Each codeword succeeds or fails independently at the frame's operating point with per-codeword loss probability p. Three delivery regimes are evaluated:

- fragment-aware partial recovery (the primary regime for the feature message): the receiver reconstructs the tensor from the codewords that arrive and zero-fills the elements carried by codewords that do not;
- packet loss: the same rule applied at packet granularity, a post-freeze transport sensitivity;
- message all-or-nothing: the message is usable only if all of its codewords decode, a pessimistic sensitivity arm for the feature message.

The delivery semantics of the two cooperative actions are not the same, and the distinction is load-bearing:

> Late fusion uses message-level delivery. For frame t, all $N_{\mathrm{cw},L,t}$ codewords must be received for the collaborator box message to be available, giving $q_{L,t} = (1-p)^{N_{\mathrm{cw},L,t}}$; otherwise execution falls back to the ego-only result. In contrast, the primary feature-fusion regime performs codeword-level partial recovery and does not fall back to ego-only when only part of the feature message is lost.

D. Effective Task Utility under Channel Corruption

Let F_t^s denote the realised frame F_1 when action s is executed at frame t. For the object-level action the message-level rule above gives

$$\begin{aligned} F_t^L(p) &= q_{L,t}\, F_t^{L,\mathrm{clean}} + \big(1 - q_{L,t}\big) F_t^{\mathrm{ego}}, \\ q_{L,t} &= (1-p)^{N_{\mathrm{cw},L,t}}, \end{aligned} \tag{2}$$

where F_t^{ego} is the ego-only frame F_1. For the feature-level action under partial recovery there is no such mixture, because a partly damaged tensor is still fused: $F_t^F(p)$ is measured on the reconstructed tensor at each sampled p and interpolated piecewise-linearly in raw p between the sampled nodes, with the endpoints locked to the clean measurement at $p = 0$ and the all-lost measurement at $p = 1$. No monotone correction is applied, because some masks raise a frame's F_1 by removing false positives and smoothing that away would erase a measured effect (Section VI-D).

Payload accounting is deliberately kept separate from this expectation: communication is charged for the message that is attempted and is never multiplied by the success probability. A failed delivery does not refund symbols already spent on the channel.

E. Communication Cost

Payload is a per-frame quantity, not a constant per action:

$$B_{t,s} = \begin{cases} 0, & s = E \text{ (or no collaborator)}, \\ B_{L,t}, & s = L, \\ B_F, & s = F, \end{cases} \tag{3}$$

in channel-use-equivalent mega-symbols (Msym). The feature payload follows from the codeword count rather than from $(\text{info} + \text{header})/R/\log_2 M$, because the direct route ignores codeword padding: $739,200$ elements at 8 bits give $5,913,600$ information bits, which packetise into 740 packets carrying $236,800$ header bits, hence $N_{\text{cw}} = 12,567$ codewords and $B_F = 3.14175$ Msym per frame. The direct route would have given 3.0752 Msym, a difference that is pure padding. The object-level payload is charged through the identical chain from each frame's own box count, at 184 bits per box, ranging from 0.00050 to 0.00525 Msym, i.e. 2 to 21 codewords. The collaborator transmits an average of 24.08 boxes per frame, producing 10.10 LDPC codewords and 0.00253 Msym. The post-fusion late-detection output contains 28.40 boxes on average, but this value is not used for payload accounting.

The mean cost over T frames is $\bar{B} = (1/T)\sum_t B_{t,s_t}$.

F. Task–Communication Objective

Let z_t be the ego-side cue vector and $(\hat{\gamma}_t, c_t)$ the estimated channel state. The selector is a map

$$s_t = g(z_t, \hat{\gamma}_t, c_t). \tag{4}$$

G. Derivation: Constrained Optimisation and Rate–Distortion View

The design problem is

$$\max_g \ \mathbb{E}[F_t^{s_t}] \quad \text{s.t.} \quad \mathbb{E}[B_{t,s_t}] \le \beta B_F, \tag{5}$$

where the budget is expressed normalised to the feature payload. We retain the three budget tiers $\beta \in \{0.10, 0.20, 0.30\}$, i.e. 0.31418, 0.62835 and 0.94252 Msym per frame; the primary cell is $\beta = 0.20$. The budget is a mean constraint across frames, not a per-frame cap—under a per-frame cap any $\beta < 1$ would make F unselectable on every frame and the problem would degenerate.

Introducing a multiplier $\lambda \ge 0$ gives the pointwise relaxed optimum

$$s_t^\star(\lambda) = \arg\max_{s \in \mathcal{S}} \big(F_t^s - \lambda B_{t,s}\big), \tag{6}$$

with ties broken $E \succ L \succ F$. The learned selector is trained against Eq. (6), and the constrained problem is addressed by sweeping λ over the family.

In rate–distortion terms, Eq. (2) and the measured $F_t^F(p)$ express a channel-induced distortion that grows with the loss rate, while payload is the rate. The selector therefore trades rate against rate-conditional distortion as a function of channel state—joint source–channel structure expressed at the granularity-action level rather than at the symbol level. This yields the design principle the paper rests on: the most informative message is not always the most useful message under communication constraints.

IV. Proposed Method

A. Overview

Instead of transmitting a fixed representation on every frame, the ego constructs an ego-local cue vector, reads its channel estimate, runs a lightweight selector, signals the chosen granularity with the 2-bit request, and the collaborator complies. Fig. 1 shows the data flow as implemented, including which quantities exist before the request is issued.

B. Ego-Side Online Task Cues

The cue vector is $z_t \in \mathbb{R}^{23}$ under the frozen schema v2_ego_local_23d: 19 statistics of the ego's own LiDAR sweep (point counts, range statistics, near/mid/far and per-sector occupancies, and three range densities), the ego's own detected box count, a binary collaborator-availability flag, and two channel fields. The complete field list with a code location per row is in the supplementary material.

Every cue is computed from the ego vehicle's own observation and its own detection stream, and every field is computable before the request is issued. This is a direct consequence of the receiver-driven architecture: nothing in z_t depends on a message the ego has not yet asked for. Ground truth, task metrics, oracle labels and delivery outcomes are excluded from the schema by construction, and the cue schema is verified against its code-defined sources.

C. Channel Quality Cue

The channel state is the estimated SNR $\hat{\gamma}_t \in \mathbb{R}$ in dB and a binary channel-type indicator c_t. In a deployed V2X stack $\hat{\gamma}_t$ comes from pilot-symbol estimation already produced by 802.11bd or NR sidelink receivers [3] and c_t can be inferred from link-layer multipath indicators. Both are treated as estimates, and the results below are conditional on their being available at the ego.

The role of $(\hat{\gamma}_t, c_t)$ differs from that of the task cues. Task cues estimate whether the frame needs richer information; channel cues estimate whether such information can be carried. A difficult frame does not by itself justify a feature request.

D. Message Branches

The object-level branch transmits the collaborator's detected boxes and is charged per frame from its own box count. The feature-level branch transmits the encoder's bottleneck tensor quantised to 8 bits per element. Quantisation is part of the transmitted form, so its cost is measured rather than assumed: on a 220-frame audit subsample of the development split the mean frame F_1 moves from 0.88808 (float) to 0.88828 (int8), a difference of +0.00020. This is a quantisation audit on that subsample and is not a clean-channel result for the full split.

E. Channel-Aware Semantic Granularity Selector

The policy $g(\cdot)$ is instantiated as a random forest [23] with 400 trees and a minimum of 2 samples per leaf, operating purely on tabular cues. It adds no learnable parameters to the perception backbone and requires no retraining of the detector, which is why a tabular learner is a natural fit; the forest is one realisation of the selection layer, not the contribution.

The selector is trained to imitate the utility-maximising action of Eq. (6), which uses realised task utility and is therefore not deployable; it supplies training labels only.

Feasibility: Mainline feasibility is structural. An action is feasible iff a collaborator is available and a transport response is defined; on a frame with no collaborator the executed action is forced to E. We do not apply a message-level BLER feasibility mask because the quantity it thresholds is the per-codeword erasure probability rather than the probability that the feature message is unusable, and because channel impairment is already reflected in the partial-recovery effectiveness. The feature output is defined and measured at every loss rate, including $p = 1$. No $F_t^F - F_t^{\text{ego}}$ safety threshold is added either: with $\lambda \geq 0$ and $B_F > 0$, $F_t^F \leq F_t^{\text{ego}}$ already implies $U_F \leq U_E$, so the objective handles it and an extra threshold would only add a researcher degree of freedom.

F. Receiver-Side Fusion and Detection

The ego fuses the received message with its local feature, $\hat{Y}_t = D\big(\Psi(F_{\text{ego},t}, m_t^{s_t})\big)$, where Ψ is the fusion module and D the detection head. If $s_t = L$ the receiver performs object-level fusion; if $s_t = F$ it dequantises the received tensor—zero-filling the elements whose codewords were lost—and performs intermediate fusion.

A zeroed collaborator tensor is not the same input as no collaborator at all. The fusion module attends over both branches, so a zero tensor still participates and dilutes the ego features. Measured rather than assumed, this is quantified in Section VI-D.

V. Experimental Protocol

A. Dataset and Implementation

We evaluate on the OPV2V dataset [1] using the OpenCOOD codebase, with a PointPillars BEV backbone [20]. One checkpoint serves all three actions, and the field of view and ground truth are identical across them, so an L-versus-F comparison is a comparison of granularity rather than of two separately trained pipelines. Exactly one collaborator is used, selected by a fixed rule, so the payload of a cooperative action is well defined. Table I states the three splits and their roles.

Evaluation is deterministic: the per-sample point shuffle is seeded on (split, scene, frame, cav), and two independent processes reproduce a run bit for bit. This is verified by a reconstruction-consistency bridge rather than assumed.

B. Message Construction and Payload Accounting

Payload is measured, not declared, and the derivation is given in full in Section III-B and expanded bit by bit in the supplementary material. Two properties matter for reading the results. First, B_F is a constant 3.14175 Msym while $B_{L,t}$ varies per frame with the collaborator's box count, so the payload axis is not a two-point scale. Second, payload is charged for the attempt (Section III-G), which is what makes an undelivered feature message expensive rather than free.

C. Channel Settings

The estimated SNR is swept over an 11-point grid and crossed with two channel types, giving the full deterministic grid of frames $\times$ SNR $\times$ channel type reported in Table I. Transport is characterised at 8 per-codeword loss rates with 4 independent channel realisations each, and the message regime is evaluated by 200 Monte-Carlo replays.

D. Selector Training and Freezing

Model selection is done on the development split only, in this order.

1) Scene-level 9-fold leave-one-scene-out (LOSO) on validate. Folds are scene-disjoint, because frames within a scene are consecutive samples of the same traffic and are far from independent.
2) Candidate and penalty selection. A candidate walk over selector configurations and penalties λ is run under the budget constraint, scored by LOSO.
3) Refit and freeze. The selected configuration—candidate 67 at $\lambda = 0.2$—is refitted on the full development split and frozen, together with the comparator: an SNR threshold at $\tau = 16.5$ dB, the best budget-feasible threshold at the primary cell. Per-tier thresholds are 18.5, 16.5 and 14.5 dB for $\beta = 0.10$, 0.20 and 0.30.
4) One-shot held-out evaluation. Test and Culver-City are evaluated only after model and comparator freezing.

LOSO is the training-time selection procedure and the scene-level bootstrap of Section V-F is the evaluation-time interval procedure; they operate at different stages and neither replaces the other.

TABLE I
Experimental protocol. The development split is where the candidate, the penalty λ and the comparator threshold were chosen; both held-out splits were evaluated only after model and comparator freezing.

split	frames	scenes	grid rows	no-collab. rows	role
validate (dev.)	1,980	9	43,560	0	development: candidate, λ and τ chosen here
Test	2,170	16	47,740	2,618	held out; evaluated only after freezing
Culver-City	550	4	12,100	1,584	held out; evaluated only after freezing (domain shift)

E. Baselines

Fixed E, Fixed L and Fixed F bound the action space. The internal decision rules are the SNR-threshold rule and two- and three-scalar hand rules over the same cues, fitted through the identical LOSO walk and budget constraint. The budget-blind oracle selects the utility-maximising action per row and is an upper-bound reference, not a deployable policy. The frozen $\tau = 16.5$ is the primary comparator because it is the only arm sharing CA-TOSG's complete payload definition.

Where2comm is included as an external feature-content baseline [5]. It is evaluated under the same data splits, field of view, ground truth and detection metrics. We report its native communication rate because the evaluated implementation transmits floating-point selected features and does not execute the locked int8 quantisation path. Consequently, it is excluded from bit-level budget-matched claims. No adaptive-policy baseline met the reproducibility bar for inclusion as a policy comparator.

F. Evaluation Metrics

AP@0.5 and AP@0.7 characterise the fixed perception arms, transport responses and the external Where2comm reference. The confirmatory comparison between the frozen selector and its budget-feasible comparator uses scene-equal mean per-frame F_1, with scene-level bootstrap intervals. Communication cost is the mean per-frame channel use defined in Section III-B.

The non-inferiority margin is $\delta = 0.005$ in absolute mean per-frame F_1. It is a preregistered engineering tolerance—how much accuracy may be given up in exchange for a substantive communication reduction—fixed on 2026-08-09, before any evaluation on either held-out split. It is not a safety-certified threshold, and no statistical or regulatory standard is claimed for its value. The confirmatory unit is the scene, not the frame. The headline statistic is therefore the scene-equal mean F_1, and intervals are scene-level bootstraps with 16 Test scenes and 4 Culver-City scenes, covering 47,740 and 12,100 evaluated rows. Every payload figure is reported under two accountings—over all frames, and over frames where a collaborator is available—and the two are never mixed.

VI. Results

A. Fixed-Action Performance and Communication Cost

Table II gives the complete arm set. On the development split the three fixed actions separate as expected in accuracy—ego-only at scene-equal F_1 0.82567, object-level at 0.85854, feature-level at 0.86506—but not at all in cost: Fixed L spends 0.00253 Msym per frame against 3.14175 Msym for Fixed F. In detection terms over the same split, ego-only reaches AP@0.5 0.72055 and object-level 0.88851, while the int8 feature message under clean delivery reaches 0.86994.

The cost asymmetry is the whole setting. Feature-level fusion buys +0.00652 in scene-equal F_1 over object-level fusion, and pays $1,244\times$ the channel use for it. That ratio, not the accuracy gap, is why a per-frame granularity decision is worth making at all.

B. Primary Held-Out Comparison

Table III and Fig. 2 give the primary cell, $\beta = 0.20$. On Test the realised payload falls by 99.85% and on Culver-City by 99.70%; on both splits the scene-level bootstrap lower bound on ΔF_1 lies outside the preregistered margin of -0.005, so accuracy non-inferiority is not established.

Both halves are stated together throughout. The saving is large—a factor of 669 and 339—and the shortfall on the bound is 0.0024 (Test) and 0.0004 (Culver-City). The criterion is defined on the bound, not on the point estimate: on both splits the point estimate lies inside the margin and the bound does not, and the bound is what it tests.

The Culver-City interval rests on only 4 scenes and is correspondingly coarse; it reproduces the shape of the Test result on much less scene diversity rather than independently corroborating it. Under the second accounting—frames where a collaborator is available, which is 94.52% of Test rows and 86.91% of Culver-City rows—the relative saving is unchanged, so none of it is an artefact of collaborator unavailability.

C. Action Distribution

The frozen selector requests E on 40.3% of Test rows and L on 59.7%, and never requests F ($\rho_F = 0.000$); the Culver-City mix is 40.7% and 59.3%. Section VII explains why the third figure is a property of the sampled penalty grid.

All three budget tiers select the same candidate and the budget never binds. At $\beta = 0.10$, 0.20 and 0.30 the frozen policy spends 0.00131, 0.00131 and 0.00131 Msym for scene-equal F_1 0.86050, 0.86050 and 0.86050—three identical operating points, because the tightest ceiling 0.31418 Msym is still $240\times$ what the policy actually spends. There are therefore not three distinct budget

TABLE II
All evaluated arms. Development-split rows are where every choice was made; the held-out rows carry the two frozen arms, which are what the pre-registered held-out comparison evaluated.

arm	scene-eq. F_1	payload (Msym)	ρ_E	ρ_L	ρ_F
development split (validate) — where every choice was made					
Fixed E (ego-only)	0.82567	0.00000	1.000	0.000	0.000
Fixed L (object-level)	0.85854	0.00253	0.000	1.000	0.000
Fixed F (feature-level)	0.86506	3.14175	0.000	0.000	1.000
CA-TOSG (frozen RF)	0.86050	0.00131	0.457	0.543	0.000
budget-blind oracle	0.88321	1.76778	0.191	0.246	0.562
SNR threshold $\tau = 16.5$	0.86289	0.57329	—	—	—
best hand rule (two_scalar_snr8.0_nbox25)	0.87437	1.24792	0.000	0.603	0.397
held-out splits — the two frozen arms, evaluated after freezing					
Test: CA-TOSG	0.87679	0.00081	0.403	0.597	0.000
Test: $\tau = 16.5$	0.88178	0.54102	—	—	—
Culver-City: CA-TOSG	0.87352	0.00147	0.407	0.593	0.000
Culver-City: $\tau = 16.5$	0.87813	0.49850	—	—	—

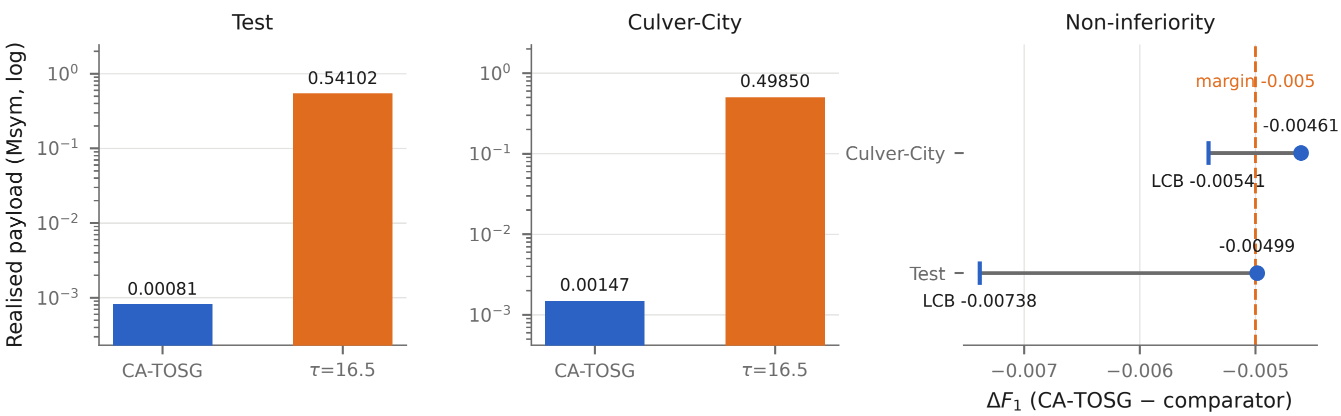

Fig. 2. Primary cell. Payload is drawn on a logarithmic axis and ΔF_1 in its own panel: the payload ratio is 669-fold while the F_1 shortfall on the bound is 0.0024, so a shared scale would erase one half of the result. The right panel puts the preregistered margin and the bootstrap lower bound on one ruler. On both splits the point estimate lies inside the margin and the bound does not.

TABLE III
Primary cell at $\beta = 0.20$. Δ is CA-TOSG minus the frozen comparator; the criterion is defined on the bootstrap bound, not on the point estimate.

	Test		Culver-City	
	CA-TOSG	$\tau = 16.5$	CA-TOSG	$\tau = 16.5$
Scene-eq. F_1	0.87679	0.88178	0.87352	0.87813
Payload (Msym)	0.00081	0.54102	0.00147	0.49850
ΔF_1	-0.00499		-0.00461	
ΔF_1 LCB	-0.00738		-0.00541	
Payload saving	99.85%		99.70%	
Scenes	16		4	

points in this study, and this is reported as a limitation of the design rather than as a robustness result.

D. Transport Mechanism

Fig. 3 and the full sweep in the supplementary material characterise the three delivery regimes.

Partial recovery is necessary, not a refinement. Under all-or-nothing message delivery the survival probability of a 12,567-codeword message is $q = 3.463 \times 10^{-6}$ even at the lowest evaluated loss rate $p = 0.001$, i.e. an expectation of 0.006857 surviving frames over the whole development split. That regime therefore measures q, not how the feature action performs, and any conclusion drawn from it would be a conclusion about message length.

Where a loss falls matters, at fixed loss amount. Conditioning on exactly equal codeword-loss counts across replicates, $\Delta F_1 \neq 0$ on 28.48% of matched pairs under fragment-aware recovery (1,731 pairs over 1,201 frames; Wilson 95% lower bound 26.40%) and on 25.57% under packet loss (704 pairs, 583 frames, lower bound 22.48%). The sufficiency rule for this test was fixed before the selection ran. We report this at the level it supports—loss position affects task performance—and do not claim the stronger form, that position matters more than amount, because no threshold for that comparison was preregistered.

Zero-tensor fusion is not ego-only inference. At $p = 1$ every codeword is lost and the fused tensor is all zeros, yet the result differs from ego-only detection in

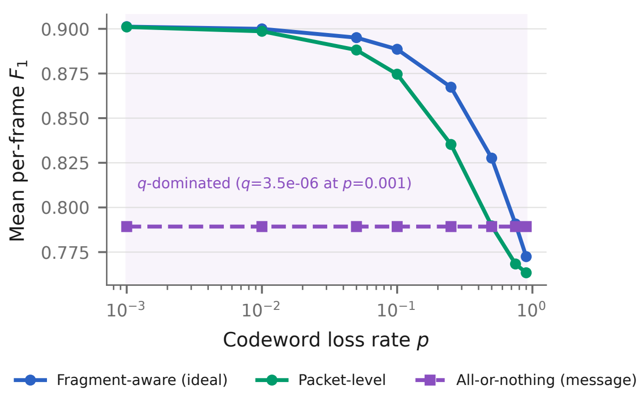

Fig. 3. Delivery regimes on the development split. Fragment-aware partial recovery degrades gracefully; all-or-nothing message delivery is governed by the survival probability q rather than by how the feature action performs.

both directions: AP@0.5 moves by +0.01195, AP@0.7 by −0.01789 and mean frame F_1 by −0.02658 (from 0.78929 to 0.76271), and the two outputs have identical boxes on 0 of the development split's frames. The claim here is inequivalence, not superiority—the signs disagree across metrics, which is precisely why the two states must not be treated as interchangeable in an evaluation pipeline.

E. External Baseline

Fig. 4 places Where2comm on its native axis. At a communication threshold of 0.02 it reaches AP@0.5 0.91708 on Test and 0.83195 on Culver-City (AP@0.7 0.82989 and 0.71588) while retaining 5.2% and 9.1% of features. The complete 24-point sweep is in the supplementary material.

The horizontal axis is the fraction of features retained, not a bit budget, and it is not converted into Msym anywhere in this paper. Doing so would require the quantisation, packetisation and coding chain that this implementation does not execute, and the resulting number would describe a system that was never run.

VII. Discussion and Limitations

Accuracy non-inferiority was not established. On both splits the point estimate lies inside the margin and the confidence bound does not. The margin was fixed in advance and the bound is what it tests.

Proportion. The saving is overwhelming; the shortfall is small. Neither may be reported without the other: stating only the saving packages a failed criterion as success, and stating only the shortfall discards a 669-fold reduction.

No three distinct budget points. All three budgets select the same candidate and the budget never binds. The contribution is therefore a granularity-control framework, not a demonstration that a learned selector beats simple rules—at equal budget it does not.

A. Why This Remains a Three-Action Method Although $\rho_F = 0$

Four measurements belong together.

First, F is genuinely the best action on 56.2% of grid rows (24,502 rows), where its realised utility exceeds both E and L by a mean net gain of 0.04890. This is not an action that never helps.

Second, the break-even penalty falls in a gap in the sampled grid. Conditional on the rows where F is best it is 0.01556; averaged over the whole grid it is 0.0048—each value carrying its conditioning set, and neither usable as a bare number. The frozen candidate carries $\lambda = 0.2$, hence $\lambda B_F = 0.62835$ Msym of penalty against a mean net gain of 0.04890—a penalty-to-gain ratio of 12.9. The preregistered grid samples λ coarsely on either side of that break-even point, so no candidate could have been selected in the region where a mixed policy exists.

Third, a finer λ scan, registered before it produced any number, shows a genuine three-action trade-off across that interval: over 19 points from $\lambda = 0.001$ to 0.019, ρ_F falls smoothly from 0.534 to 0.152.

Fourth, all of this is an exploratory diagnostic on the development split. It enters no primary criterion and the frozen candidate was not replaced on the strength of it.

Taken together, $\rho_F = 0$ is a property of where the penalty grid sampled, not evidence that the feature action is worthless.

B. Other Limitations

Culver-City rests on 4 scenes. The bootstrap resamples them, so its interval is coarse, and Culver-City is a simulated domain shift rather than real-road data.

One ego–collaborator pair. The payload of a cooperative action is defined against exactly one collaborator; a deployment serving several would have to re-derive both the payload and the budget.

A flat block-fading channel. Each codeword sees its own realisation with perfect receiver channel state information; correlated fading and imperfect estimation are not modelled.

Selector-only latency. The selector's own inference cost is measured; total end-to-end system latency, including any cost of keeping the cooperative pipeline resident, is not.

One dataset family and one detector. Every result here is conditional on OPV2V and on a single PointPillars checkpoint.

VIII. Conclusion

Three transmission granularities were placed under one detector, one field of view and one ground truth, and charged through one measured transport chain. Communication falls by a factor of 669 on Test and 339 on Culver-City relative to a budget-feasible threshold policy, while accuracy non-inferiority is not established on either. The conditions are one dataset family, one detector, a single ego–collaborator pair, and a flat block-fading channel.

The mechanism results are the part most likely to transfer. Partial recovery is not an optimisation but a precondition for studying feature-level transport at

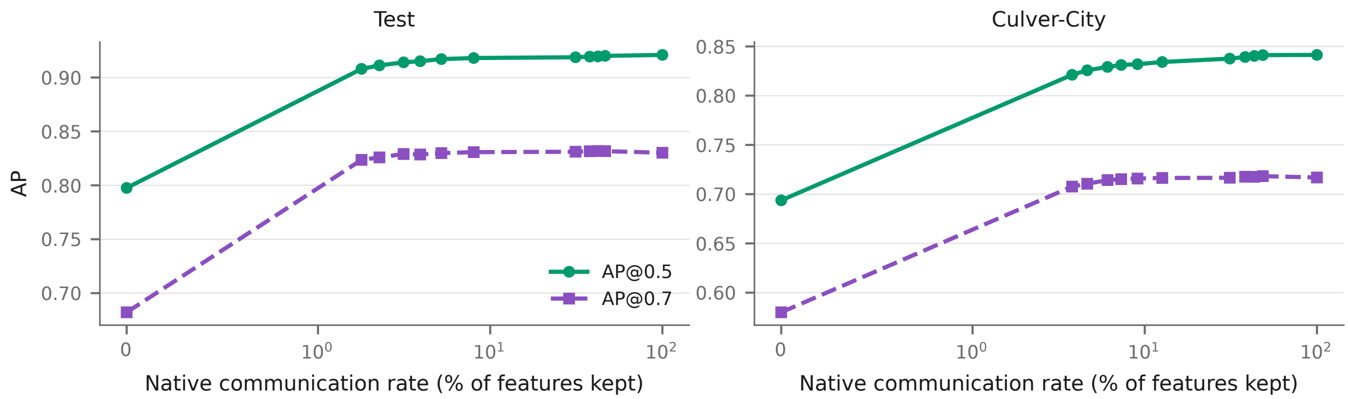

Fig. 4. Where2comm on its native communication-rate axis. The horizontal axis is the fraction of features retained, not a bit budget.

realistic message lengths; loss position changes the task outcome at fixed loss amount, so a transport model that tracks only how much was lost is incomplete; and a zeroed collaborator tensor is not an ego-only input, so the two must be distinguished in any evaluation that models delivery failure. Each of these is a statement about the transport–perception interface rather than about one policy, and each was measured under a protocol in which the detector, the field of view and the ground truth were held fixed.

References

[1] R. Xu, H. Xiang, X. Xia, X. Han, J. Li, and J. Ma, “OPV2V: An open benchmark dataset and fusion pipeline for perception with vehicle-to-vehicle communication,” in Proc. IEEE Int. Conf. Robotics and Automation (ICRA), 2022, pp. 2583–2589.

[2] R. Xu, H. Xiang, Z. Tu, X. Xia, M.-H. Yang, and J. Ma, “V2X-ViT: Vehicle-to-everything cooperative perception with vision transformer,” in Proc. Eur. Conf. Comput. Vis. (ECCV), 2022, pp. 107–124.

[3] C. Iliopoulos, A. Iossifides, C. H. Foh, and P. Chatzimisios, “IEEE 802.11bd for next-generation V2X communications: From protocol to services,” IEEE Commun. Standards Mag., vol. 9, no. 2, pp. 88–98, Jun. 2025, dOI: 10.1109/MCOMSTD.2025.3569015.

[4] 3GPP, “Study on evaluation methodology of new V2X use cases for LTE and NR,” 3GPP, Tech. Rep. TR 37.885, 2019, rel-15, v15.3.0.

[5] Y. Hu, S. Fang, Z. Lei, Y. Zhong, and S. Chen, “Where2comm: Communication-efficient collaborative perception via spatial confidence maps,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 35, 2022, pp. 4874–4886.

[6] Y. Sheng, H. Ye, L. Liang, S. Jin, and G. Y. Li, “Semantic communication for cooperative perception based on importance map,” Journal of the Franklin Institute, vol. 361, no. 6, p. 106739, 2024.

[7] J. Gan, Y. Sheng, H. Zhang, L. Liang, H. Ye, C. Guo, and S. Jin, “SComCP: Task-oriented semantic communication for collaborative perception,” IEEE Trans. Veh. Technol., 2026, dOI: 10.1109/TVT.2026.3654968.

[8] T.-H. Wang, S. Manivasagam, M. Liang, B. Yang, W. Zeng, and R. Urtasun, “V2vnet: Vehicle-to-vehicle communication for joint perception and prediction,” in European Conference on Computer Vision (ECCV), 2020, pp. 605–621.

[9] R. Xu, Z. Tu, H. Xiang, W. Shao, B. Zhou, and J. Ma, “Cobevt: Cooperative bird’s eye view semantic segmentation with sparse transformers,” in Conference on Robot Learning (CoRL), 2022.

[10] Y. Li, S. Ren, P. Wu, S. Chen, C. Feng, and W. Zhang, “Learning distilled collaboration graph for multi-agent perception,” in Advances in Neural Information Processing Systems (NeurIPS), vol. 34, 2021.

[11] Y.-C. Liu, J. Tian, N. Glaser, and Z. Kira, “When2com: Multi-agent perception via communication graph grouping,” in IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 4106–4115.

[12] Y. Zhang, H. An, Z. Fang, G. Xu, Y. Zhou, X. Chen, and Y. Fang, “SmartCooper: Vehicular collaborative perception with adaptive fusion and judger mechanism,” in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2024.

[13] Q. Xie, X. Zhou, T. Qiu, Q. Zhang, and W. Qu, “Soft actor–critic-based multilevel cooperative perception for connected autonomous vehicles,” IEEE Internet of Things Journal, vol. 9, no. 21, pp. 21 370–21 381, 2022.

[14] P. Liu, Z. Wang, G. Yu, B. Zhou, and P. Chen, “Region-based hybrid collaborative perception for connected autonomous vehicles,” IEEE Trans. Veh. Technol., vol. 73, no. 3, pp. 3119–3128, 2024, dOI: 10.1109/TVT.2023.3324439.

[15] E. Bourtsoulatze, D. B. Kurka, and D. Gündüz, “Deep joint source-channel coding for wireless image transmission,” IEEE Transactions on Cognitive Communications and Networking, vol. 5, no. 3, pp. 567–579, 2019.

[16] H. Xie, Z. Qin, G. Y. Li, and B.-H. Juang, “Deep learning enabled semantic communication systems,” IEEE Transactions on Signal Processing, vol. 69, pp. 2663–2675, 2021.

[17] D. Gündüz, Z. Qin, I. E. Aguerri, H. S. Dhillon, Z. Yang, A. Yener, K. K. Wong, and C.-B. Chae, “Beyond transmitting bits: Context, semantics, and task-oriented communications,” IEEE Journal on Selected Areas in Communications, vol. 41, no. 1, pp. 5–41, 2023.

[18] J. Shi, J. Zhao, L. Zhuo, X. Wang, X. Zhan, and H. Liu, “V2V cooperative perception with adaptive communication loss for autonomous driving,” IEEE Trans. Intell. Transp. Syst., vol. 26, no. 10, pp. 14 866–14 878, 2025, dOI: 10.1109/TITS.2025.3582601.

[19] J. Gan, L. Liang, H. Zhang, C. Guo, and S. Jin, “CoDS: Collaborative perception via digital semantic communication,” arXiv preprint arXiv:2512.22513, 2025, v1, 27 Dec 2025.

[20] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, “PointPillars: Fast encoders for object detection from point clouds,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2019, pp. 12 697–12 705.

[21] ETSI, “Intelligent transport systems (ITS); vehicular communications; basic set of applications; part 2: Specification of cooperative awareness basic service,” European Telecommunications Standards Institute, EN 302 637-2 V1.4.1, 2019.

[22] ——, “Intelligent transport systems (ITS); vehicular communications; basic set of applications; collective perception service; release 2,” European Telecommunications Standards Institute, TS 103 324 V2.1.1, 2023.

[23] L. Breiman, “Random forests,” Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001.